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Lab 6 - Linear Regression

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Part 1

1. Changing the value of standard deviation

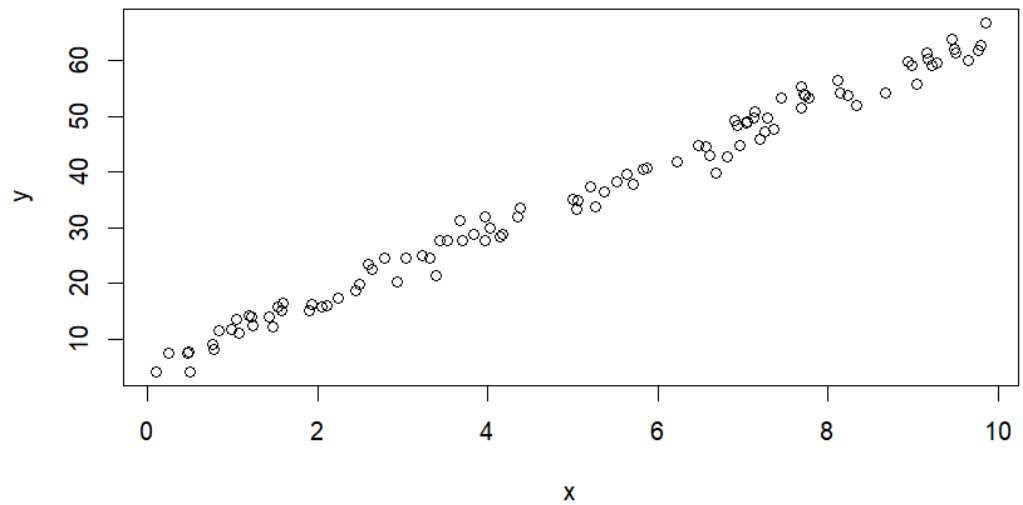

Figure 1: Scatter Plot of Data with std = 2

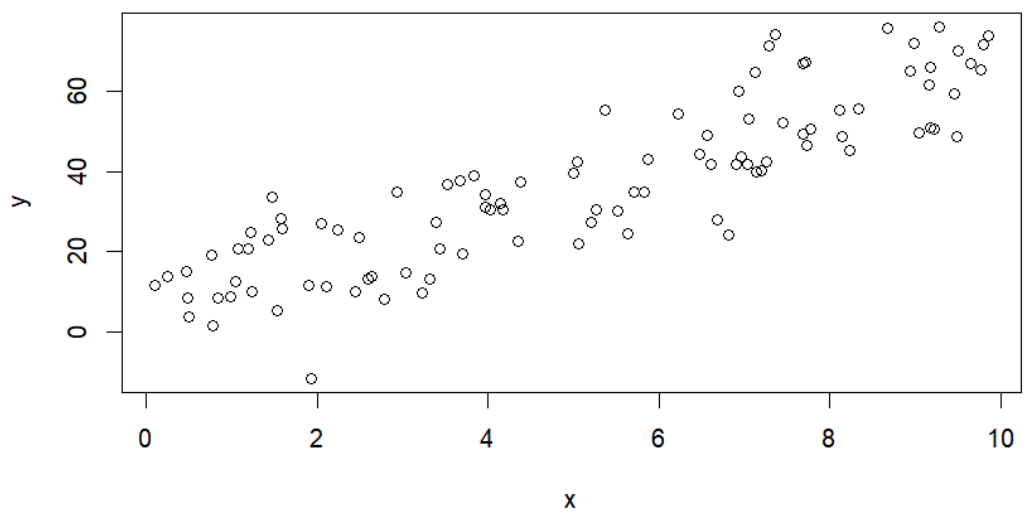

Figure 2: Scatter Plot of Data with std = 10

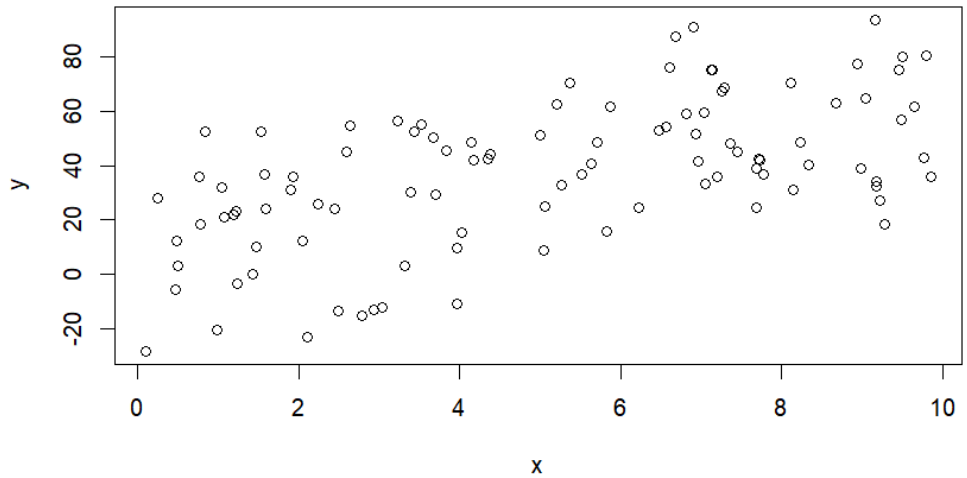

Figure 3: Scatter Plot of Data with std = 20

Conclusion:

As the standard deviation increases, the data points become more scattered.

2. Effect of Standard Deviation on Linear Model Coefficients

std	b0	b1
2	4.88517	5.997
10	8.0697	5.6281
20	10.6504	5.2338

3. Effect of Standard Deviation on R-Squared

std	R-Squared
2	0.9895
10	0.7516
20	0.3968

Conclusion:

The value of R-Squared decreasing when standard deviation (scattering) of data increases because it becomes harder for the model to find a close fit for the data in a linear way

4. Residual Plot Analysis

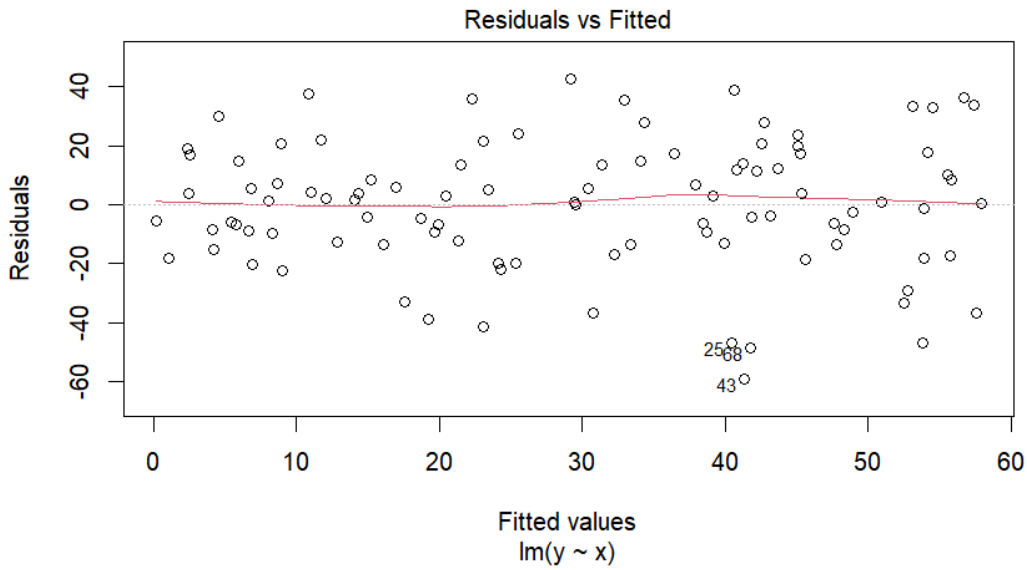

Figure 4: Residuals Plot for std = 20

Conclusion:

The residuals show no clear pattern, which is good. However, their wide range indicates larger differences between observed and predicted values, meaning the model becomes less accurate as data variance increases.

Part 2

5. Residual Plot with Non-Linear Term

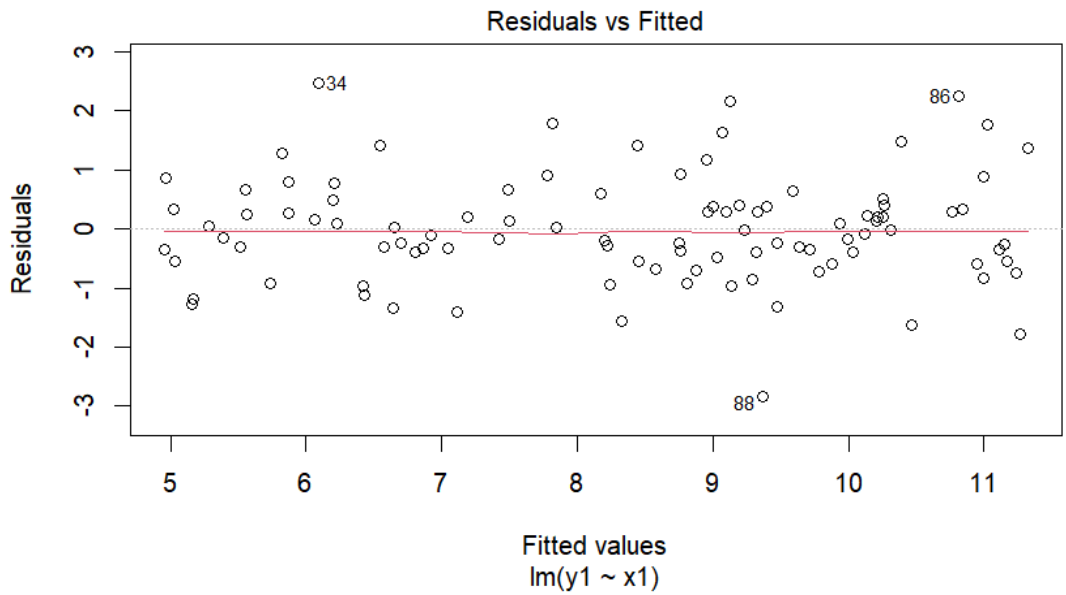

Figure 5: Residuals Plot with coefficient of $x^2 = 0.1$

Conclusion:

The residuals are good because they show no clear pattern and fall within a small range of [−2,2], indicating that the model fits the data well.

6. Changing the Non-Linear Coefficient

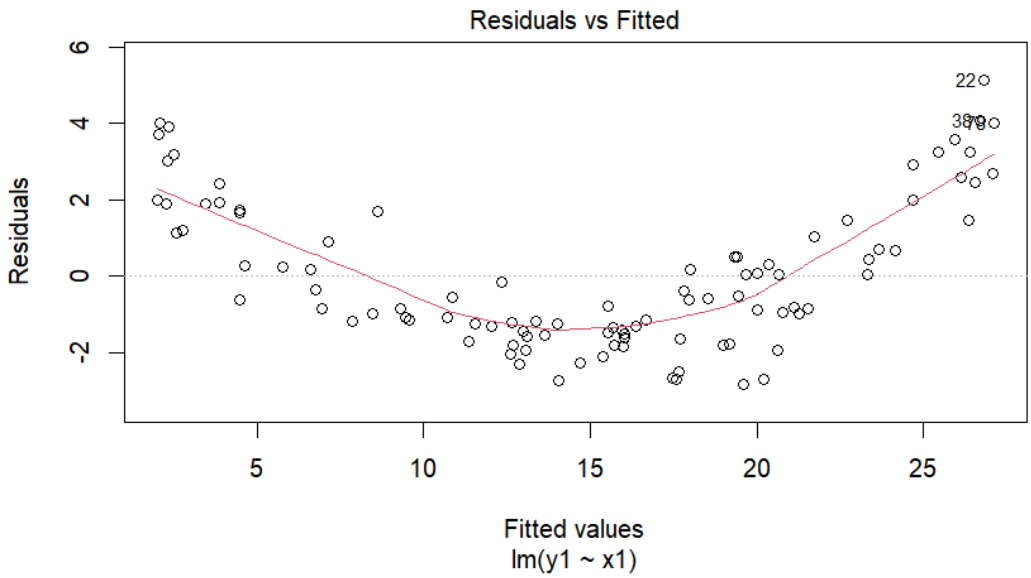

Figure 6: Residuals Plot with coefficient of $x^2 = 20.1$

Conclusion:

When the non-linear coefficient becomes large, the residual plot shows a curved pattern, indicating non-linear relationship and that a linear model is not suitable.

Part 3

7. Dataset Variables

- LungCap
- Age
- Height
- Smoke
- Gender
- Caesarean

8. Scatter Plot

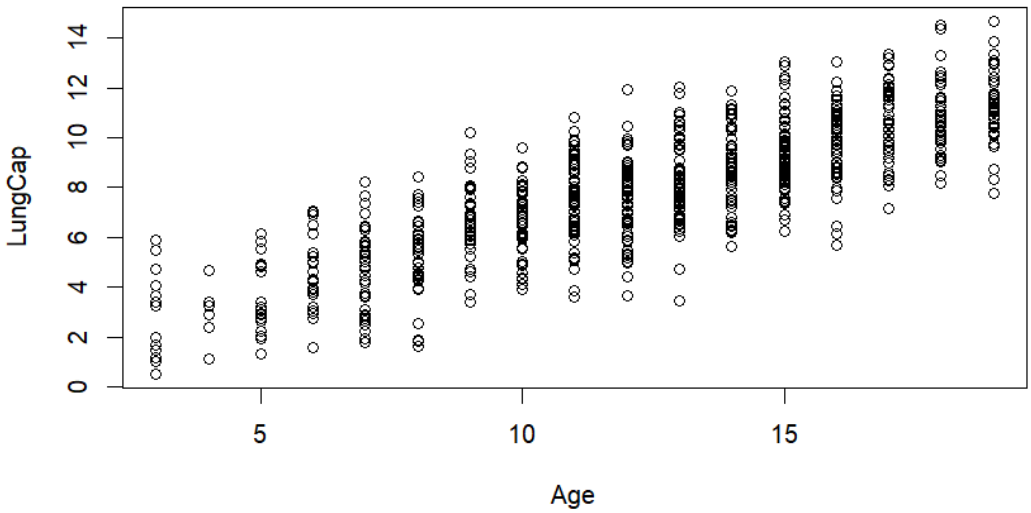

Figure 7: Scatter Plot for Age and LungCap

9. Pairwise Scatter Plot

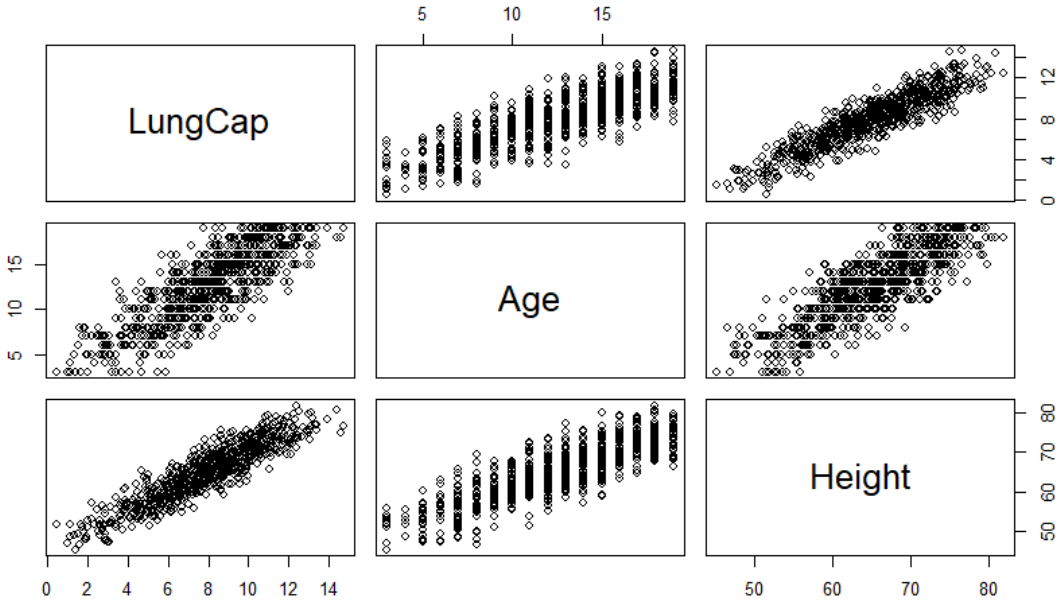

Figure 8: Pairwise Plot for Age, Height and LungCap

10. Correlation Analysis

Variable	Correlation with LungCap

Age	0.8196749
Height	0.9121873

11. Most Correlated Variable

Height shows the strongest correlation with LungCap between the two variables.

12. Relationship Between Height and LungCap

Yes, Height and LungCap are positively correlated. This can be observed from the scatter plot.

15. R-Squared Interpretation

It is 0.85 it shows that model performance is strong, data not really scattered

16. Model Coefficients Interpretation

Variable	Age	Height	Smokeyes	Gendermale	Caesareanyes
Coefficient	0.16053	0.26411	-0.60956	0.38701	-0.21422

Age, Height, and Gendermale have positive effects on the outcome, while Smokeyes and Caesareanyes have negative effects. We should check the significance of these variables and remove any that are not important to improve the model.

17. Linear Model Visualization

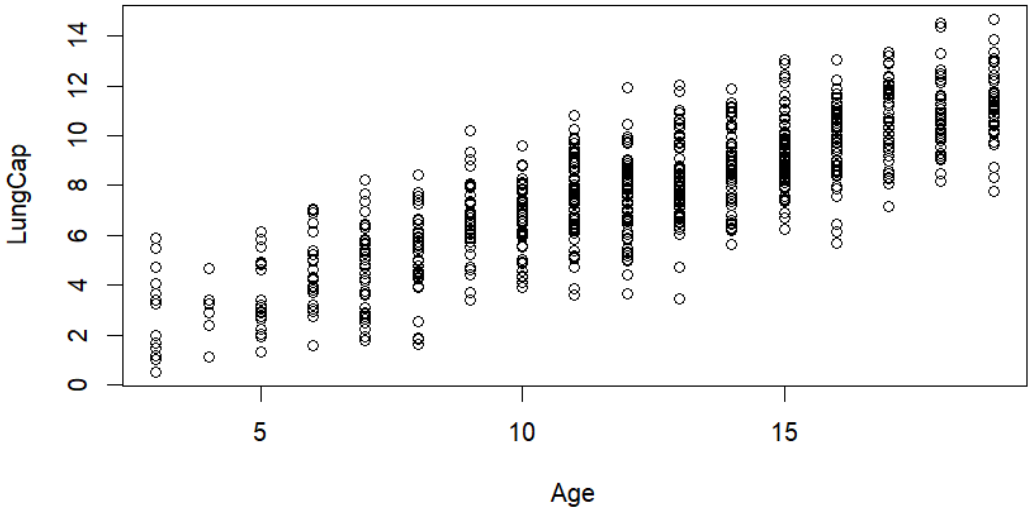

Figure 9: Scatter Plot for Age and LungCap with absline()

The regression line is not displayed correctly because the model includes multiple predictors, while absline() expects a simple linear model.

19. New Model Results

Variable	Age	Smokeyes	Caesareanyes
Coefficient	0.556	-0.64	-0.146

After modifying the model and removing some variables, the line now shows as age is the only continuous variable.

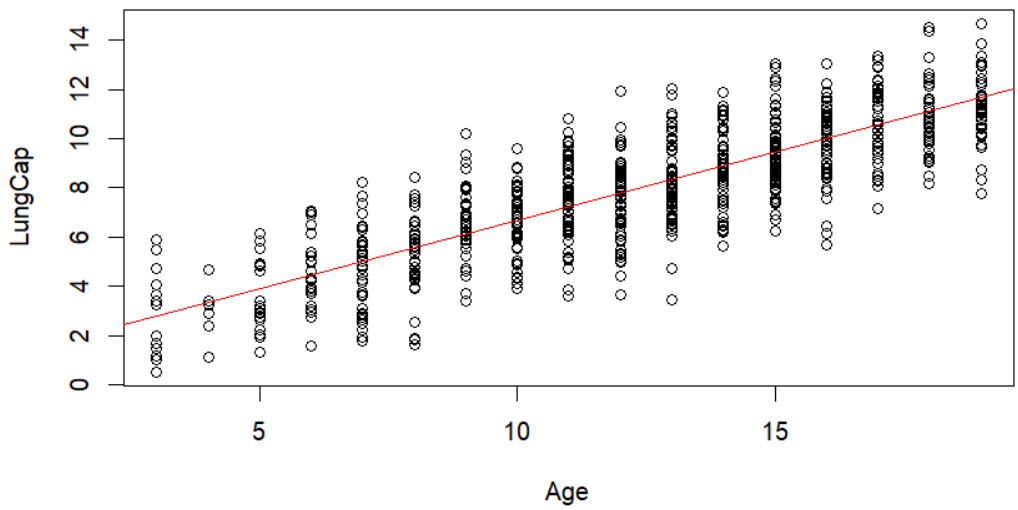

Figure 10: Scatter Plot for Age and LungCap with absline()

21. Calculate the mean squared error (MSE) of the training data.

MSE = 2.280169